

3.2 Unit 2 BIOL2 The variety of living organisms

Unit 2 builds on concepts developed in Unit 1.

Although a species may be defined in terms of similarity, there is frequently considerable intraspecific variation and this is influenced by genetic and environmental factors. DNA is an information-carrying molecule, and similarities and differences in the sequence of bases in DNA result in genetic diversity.

The variety of life is extensive and is reflected in similarities and differences in its biochemical basis and cellular organisation. Factors such as size and metabolic rate affect the requirements of organisms and this gives rise to adaptations such as specialised exchange surfaces and mass transport systems.

Classification is a means of organising the variety of life based on relationships between organisms and is built round the concept of a species. Originally, classification systems were based on observable features but more recent approaches draw on a wider range of evidence to clarify relationships between organisms. Variation that exists at the interspecific level contributes to the biodiversity of communities and ecosystems.

It is anticipated that this unit will allow opportunity for further development of the skills of application and analysis as well as for the acquisition of additional investigatory skills associated with *Investigative and practical skills*, detailed in Unit 3.
